LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

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Discipline: Mechanical Engineering

4DMedical
Industry Partner

1. Introduction

1.1 Project background

Commercial models cannot meet specific research and development needs. 4DMedical's XV scanner enables research using our controller and the medical team's lung model. This stage validates one 250 mL branch prototype.

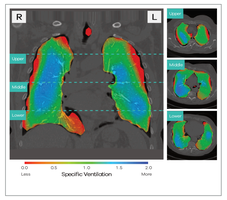


Fig. 1. XV LVAS ventilation map.

1.2 Project objective

- Adjust breathing rate and I:E.
- Control tidal volume.
- Program regional airflow.
- Support repeatable breathing cycles.
- UI for interfacing.

1.3 Nominal breathing reference

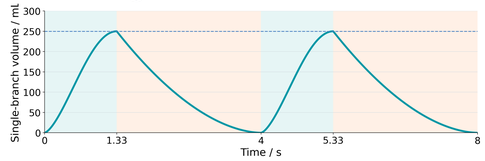


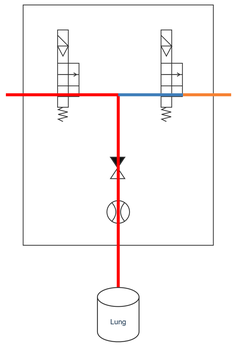
Fig. 2. Nominal breathing cycle: 15 bpm; I:E = 1:2.

Respiratory rate	**12–16 bpm; nominal 15 bpm**
Inhale:Exhale time	**1:2; 1.33 s / 2.67 s at 15 bpm**
Tidal volume	**250 mL per branch**
Regional airflow	**Independently programmable by branch**

Nominal reference values shown; operating parameters are programmable.
Source: 4DMedical, XV LVAS Info Sheet, DOC-5070 US V1 (2023), pp. 1-2.

2. Pneumatic System & Hardware

2.1 Pneumatic system prototype



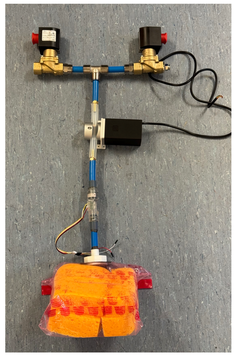
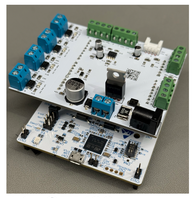


Fig. 3. Single-lung pneumatic route and 250 mL prototype.

2.2 Embedded control architecture

PC / Python UI → MCU → 24 V bridge PCB → Valves + sensor

MCU executes commands from UI and acquires flow data. A custom bridging PCB integrates 24 V valve drivers, proportional-valve control circuitry, and sensor interfaces.



Scan for more details

Fig. 4. Control architecture and bridge PCB connected to STM32.

3. Integrated Control System & Software

3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

3.2 Operator and developer interface

User Mode **Developer Mode**

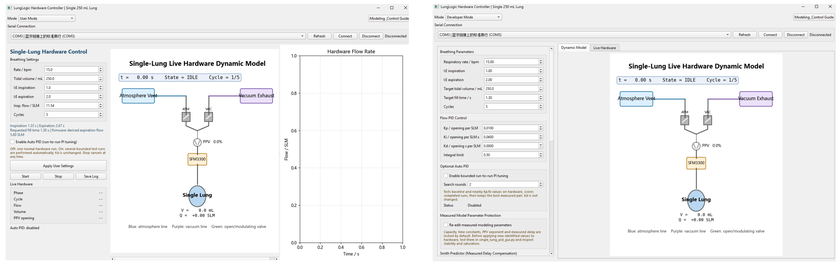


Fig. 5. User and Developer modes.

3.3 Model-assisted flow feedback

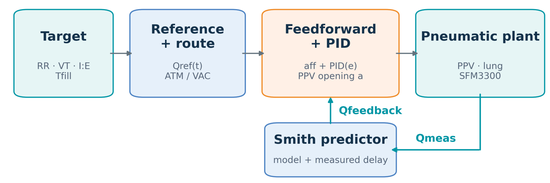


Fig. 6. Feedback control with optional Smith prediction.

3.4 Optional Auto PID and measured-delay compensation

Auto PID scores bounded Kp/Ki over full runs by flow, volume and saturation; optional Smith prediction uses the model and measured delay.

4. Results

4.1 Digital-twin simulation

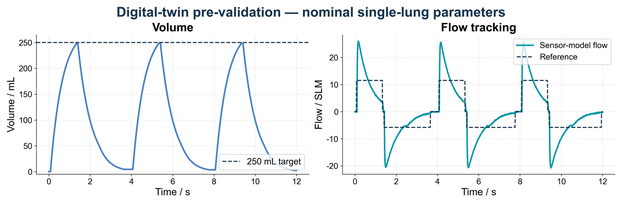


Fig. 7. Digital-twin volume and flow tracking at 15 bpm.

Physical tests refined model parameters and showed trends consistent with the model: higher PPV voltage restricts airflow and slows filling and emptying near closure. Measured responses guide phase-specific valve settings for adjustable flow and repeatable cycles. Future work will reuse this system with the medical team's biomimetic lung.

4.2 Physical measurements

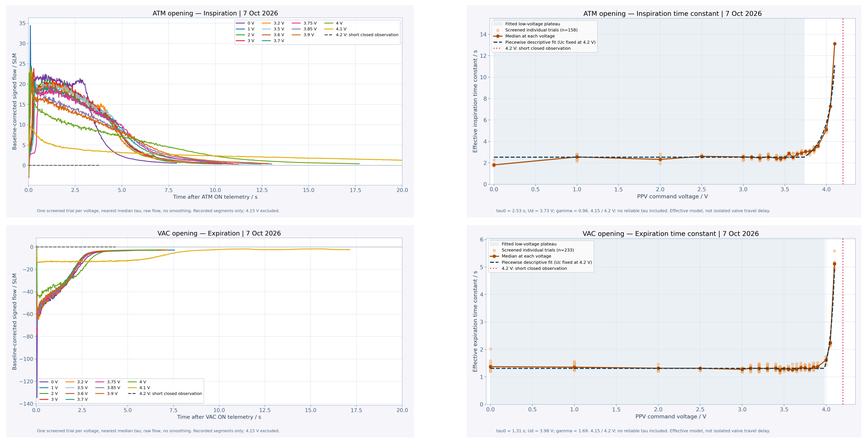


Fig. 8. Inspiratory and expiratory pinch-valve opening tests and time constants.